SAWSTAR

User Manual

Simple Synth - Big Sound
The Sounds of Trance

About this guide

SAWSTAR is a VST3 synthesizer for saw leads, pads, plucks and related sounds. This guide describes the 1.0.0 release and its 24 embedded factory presets. Edition: 11 September 2026. Screenshots show the real development GUI; preset counts and library contents can differ.

Reading the interface

Names such as MAIN, ADVANCED, PRESETS, Save As and NOISE COLOR are the labels to look for in the plugin. Start with MAIN for sound design; ADVANCED contains performance and deeper controls; PRESETS manages the library and explains saved sounds.

Contents

02 / Installation and first sound

Choose the correct package

Choose Windows x64 or native ARM64 to match your DAW. macOS Universal includes Intel and Apple Silicon. Only VST3 is supplied. Windows needs an OpenGL 2-capable driver; macOS uses Metal. Build targets are Windows 10 x64, Windows 11 ARM64 and macOS 11; these are not verified minimum-system guarantees. No separate DirectX or Visual C++ Redistributable is required.

Install

Close your DAW before replacing the plugin, and keep the previous bundle as a backup. Run the Windows Setup EXE, or open the macOS DMG and run its PKG. Administrator authorization is required. macOS installs to /Library/Audio/Plug-Ins/VST3/; Windows installs to C:\Program Files\Common Files\VST3\ . Move older user-local Mac copies outside VST3 folders to avoid duplicates. Restart the host and rescan plugins if required. Do not copy only the executable from inside the bundle.

Unsigned packages

macOS packages have no Apple Developer ID signature or notarization; Windows installers are not Authenticode signed. Security warnings may appear. Follow the package INSTALLATION guide and approve only the trusted SAWSTAR download. Do not disable system-wide protections.

First sound in REAPER

Insert SAWSTAR as an instrument on a track. Click a key on its onscreen keyboard. For an external keyboard, choose the track MIDI input, arm the track and enable input monitoring as appropriate. Check that the track and master are unmuted and routed to your audio device. Select SuperSaw One in the top preset selector and play a few notes.

Updating safely

Back up user presets and important projects before updating a development build. Check the About build information after restarting the DAW: an open host may retain the old plugin. Factory presets are embedded and need no separate bank installation.

03 / Navigation and controls

Three views, one instrument

MAIN, ADVANCED and PRESETS share the top preset selector and the bottom keyboard, wheels and status strip. Switching views does not load a different sound. The preset selector and its arrows change the loaded preset. The gear menu provides interface scaling, the downloads link and About; OK closes About.

Editing a sound

Drag knobs and faders to change values. Use the value-entry field where available for an exact number; confirm the entry with Enter. Dropdown controls choose an option. The blue active state distinguishes enabled switches from their darker inactive state. Changing a control affects the sound immediately; save the sound if you want to keep it.

Keyboard and wheels

The compact keyboard covers MIDI notes 36-96; the host can send the full MIDI note range. There is no octave-switch control for the onscreen keyboard. PITCH springs back to center when released. MOD retains its position. Double-click either wheel to reset it. The GUI keyboard and wheels use MIDI channel 1; external MIDI controller processing is channel-aware.

What is saved

Sound parameters belong to presets and DAW state. Wheel positions are temporary MIDI controller values: record wheel movements as MIDI in your DAW. The footer reports sample rate, active voices and CPU information. These are status indicators, not fixed performance promises.

04 / Sound sources and output

OSC 1 and OSC 2

Choose Saw, Square, Triangle or Sine in the waveform selector. OCT transposes the oscillator by octaves. DETUNE spreads the side oscillators. UNISON blends six detuned side oscillators with the central oscillator; it is not a polyphony selector. At zero, the center dominates. WIDTH spreads the unison across stereo. Start with a saw, moderate UNISON and a small DETUNE value, then adjust by ear.

SUB and MIXER

SUB offers Sine, Triangle and Square with a lower octave range. Its volume is controlled by the SUB mixer fader. The four mixer faders balance OSC 1, OSC 2, SUB and NOISE. A waveform can be configured but remain inaudible if its mixer level is zero.

Noise

Choose White, Pink or Dark noise below the mixer. White is bright and broad, Pink emphasizes lower frequencies relative to White, and Dark provides a darker character. NOISE COLOR further shapes the tone: negative values darken it and positive values brighten it by reducing low-frequency content. The NOISE fader controls how much noise enters the sound.

OUTPUT

VOLUME sets the master output level; BOOST adds gain. Watch the output meter and listen while balancing loud presets. WIDE and its AMOUNT expand existing stereo separation after the effects; they do not create stereo from a mono signal. Oscillator WIDTH works earlier in the signal path. Check mono compatibility when using wide sounds. Internal peak protection does not guarantee that later DAW effects or summed tracks will never clip.

05 / Filter, envelopes and scope

FILTER

Choose Low Pass 12, Low Pass 24, High Pass 12 or Band Pass 12. CUTOFF sets the filter frequency, RES emphasizes the cutoff region and DRIVE adds filter character. FILTER MIX blends the filtering into the sound: at zero, a cutoff change may be inaudible. KEY TRACK makes cutoff follow played pitch. The cutoff display is an illustration of the filter setting.

Two envelopes

AMP ENV controls the loudness of each note. A is attack time, D is decay time, S is the held sustain level, and R is release time after letting go. FILTER ENV has its own ADSR and AMOUNT: positive amount opens the filter, negative amount moves it in the opposite direction. A short decay and low sustain are useful for plucks; longer attack and release suit pads.

Realtime waveform

The green scope below AMP ENV shows the summed voice signal before effects, master width and output gain. Its horizontal axis is time; the center line is zero amplitude, not 0 Hz. The dark grid and triggered display help reveal waveform structure. The display scales amplitude automatically, so it is not a calibrated loudness meter.

Interpreting the trace

A single periodic note is easier to read than a detuned chord or noise. A complex moving trace is normal for those sounds. Stereo is combined for this display, so cancellation can reduce the trace. Changing OUTPUT or adding delay will not change this pre-effects view. The scope is visual only and does not process the audible signal.

06 / Effects

Signal order

The voice signal reaches Chorus, then Delay, then Reverb, followed by master stereo width and output processing. The MAIN FX section provides the quick controls; open the corresponding effect on ADVANCED for detailed settings. Enable an effect and raise its MIX to hear it.

Chorus

Chorus adds movement and thickness using a modulated delayed signal. MIX balances wet and dry sound. RATE controls movement speed and DEPTH its extent. For a steady lead, use a modest mix and slow rate; excessive depth can make pitch movement more obvious.

Delay

MIX sets the echo level. TIME selects milliseconds in free mode; tempo sync uses a musical DIVISION and the host tempo. FEEDBACK sets the number and persistence of repeats, while TONE darkens or brightens them. Stereo and Ping Pong modes give different spatial patterns. Lower feedback and mix if echoes obscure the next phrase.

Reverb

MIX adds the reverberant signal. SIZE changes the space character, DECAY controls tail duration and DAMPING shapes the high-frequency decay. Use shorter, quieter reverb for rhythmic plucks and more space for pads. Long release, delay feedback and reverb tails can all continue after a key is released; this is not automatically a stuck-note fault.

07 / Performance and arpeggiator

Voice modes and glide

Poly plays chords with up to 16 synth voices. Mono and Legato are single-note performance modes; Legato suits connected phrases. GLIDE controls the transition between pitches. Its mode can apply it always or only when notes overlap. PITCH BEND range is adjustable up to 24 semitones; MOD DEPTH controls the built-in modulation-wheel route to cutoff.

Build an arpeggio

Enable ARP and hold a chord. ORDER chooses Up, Down, Up/Down, Random or Played. Played follows key order. RATE selects the musical step length; OCTAVES extends the pattern across 1-4 octaves. GATE controls how much of a step the note lasts. SWING alternates longer and shorter steps.

HOLD

HOLD latches the chord after your fingers leave the keys. After all physical keys are released, playing a new chord replaces the latched phrase. Disable HOLD to stop relying on the latch; sustain-pedal behavior can still affect held notes. Disable ARP to return to ordinary playing.

Tempo behavior

The first held key starts the phrase. The arp follows host BPM but is not an absolute song-grid sequencer. It can play while the host is stopped; stopping a running host clears the phrase, and a new key starts it again. Generated notes drive SAWSTAR internally; the arp is not presented as a MIDI-output or programmable step-sequencer feature.

08 / LFO and modulation

Two independent LFOs

LFO 1 / LFO 2 selects which LFO you edit; both can operate simultaneously. Choose Sine, Triangle, Ramp or Square. RATE uses free frequency or tempo-synced division. Retrigger restarts the phase on the first key; free phase keeps it moving. The direct destination can be cutoff, pitch, amplitude or pan. AMOUNT sets the direct route depth.

Four modulation rows

Each row selects a source, a destination and a signed amount. Sources are Off, LFO 1, LFO 2, Mod Wheel, Velocity and Aftertouch. Destinations include Filter Cutoff, Pitch, Amp Level, Pan and Noise Color. Negative amount reverses the direction. Aftertouch means MIDI channel pressure; polyphonic key pressure and MPE are not supplied here.

Avoid accidental double modulation

An LFO matrix source is taken before its direct AMOUNT. Therefore, setting direct AMOUNT to zero does not silence an active matrix route. Multiple routes add together. Likewise, MOD DEPTH can still move cutoff while a separate Mod Wheel matrix row is enabled. Set MOD DEPTH to zero if the matrix should provide the only wheel assignment.

A subtle pad example

Select a pad. Set LFO 2 to a slow sine, with direct AMOUNT zero. In one matrix row choose LFO 2, Pan and a small positive amount. Listen in stereo, then reduce the row amount to compare. Save As preserves the result as a new user preset.

09 / Preset workflow

Browse and understand

Click a preset row to load it. Category and search filters narrow the list without changing the sound by themselves. Search matches names, categories and tags. The heart marks a favorite. HOW IT WORKS shows the saved preset settings, not your unsaved edits; it updates when the edited sound is saved.

Save and Save As

Save overwrites the active changed user preset after confirmation. Factory presets and templates are read-only: use Save As to create a user copy. Save As opens a naming/file dialog and creates a new .sawstar file; use a different name when a file already exists. An externally saved file is also copied into the managed library when possible. A collision can prevent that library copy.

Import, Copy and Paste

Import accepts multiple .sawstar files at once, so a collection need not be imported one file at a time. It reports skipped or failed items and does not automatically play a preset. Copy/Paste transfers sound settings within the editor and creates no file until saved. This clipboard is temporary and is cleared when the editor closes.

Rename, Delete and Initialize

Rename changes a selected user preset name. Delete asks for confirmation and retains a .deleted backup; it does not stop the currently playing sound. Initialize asks before resetting the sound to Init. Save changes you want to keep before initializing or loading another sound.

Library and recovery

User files: macOS ~/Library/Application Support/SAWSTAR/Presets ; Windows %APPDATA%\SAWSTAR\Presets . Save backups are kept in .sawstar-backups below that folder. Restore a backup by copying it into the library under an unused .sawstar filename. DAW projects store sound parameters independently, but may not restore the association with a user file for Save; reload that user preset to re-establish it.

10 / Factory library: 24 starting points

Seven Templates

Init • Pad-Init • Lead-Init • Pluck-Init • Bass-Init • Keys-Init • Arp-Init. These are deliberate starting points rather than finished demonstrations. Effects and movement are kept off. In Arp-Init, enable ARP yourself to begin developing a pattern.

Leads

Wide Saw Lead • SuperSaw One • Silk Lead • Solid Saw Lead. Explore stereo unison, fuller saw layers and connected lead playing. Compare with Lead-Init to understand which changes create the finished character.

Pads

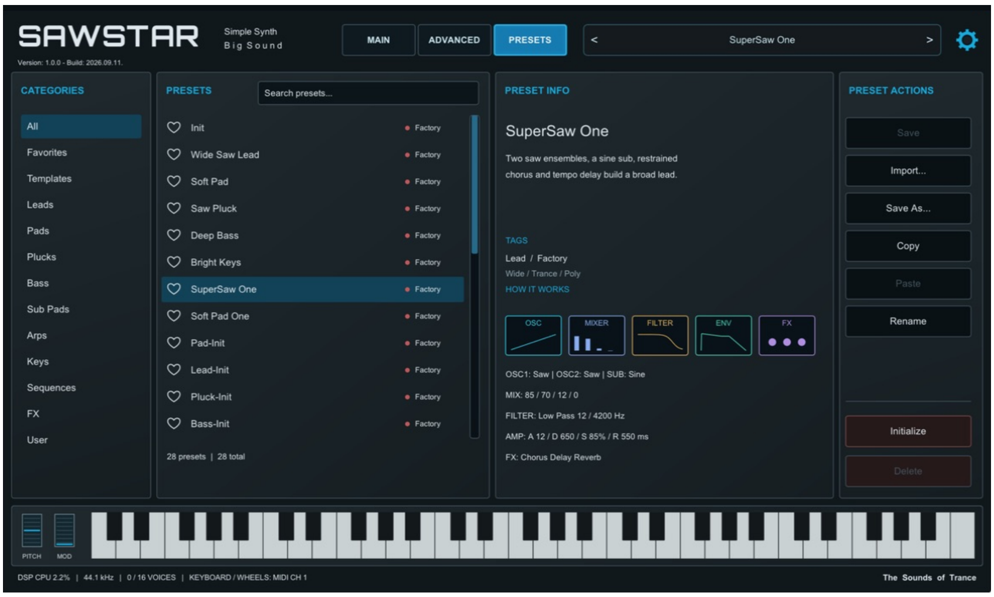
Soft Pad • Soft Pad One • Warm Horizon • Velvet Pad. Use held chords to hear envelope development and tails. Keep space for other instruments when adding width and reverb.

Plucks, basses and keys

Plucks: Saw Pluck • Glass Pluck • Trance Pluck. Basses: Deep Bass • Round Sub. Keys: Bright Keys • Air Keys. Try short and long notes, plus different velocities, to understand the envelope response.

Arpeggiated sounds

Simple Steps • Octave Steps. Hold a chord and explore rate and gate before changing the oscillators. Empty categories in the browser do not indicate a missing installation; this edition contains the 24 named presets above. User presets are additional and are not part of the embedded factory count.



11 / Three sound design exercises

A broad lead

Load Lead-Init. Choose Saw for OSC 1, raise UNISON gradually and use moderate DETUNE. Bring in OSC 2 at a lower mixer level to add body. Adjust WIDTH while checking mono. Use a fast amp attack and a release that does not overlap phrases too much. Add a restrained delay, then balance OUTPUT against another instrument. Save As with your own name.

A soft pad

Load Pad-Init. Blend two oscillators and lengthen amp attack and release. Lower cutoff with FILTER MIX active, and use a gentle filter-envelope amount. Add a small chorus and reverb. If needed, introduce a very slow, low-depth pan or cutoff LFO. Avoid adding modulation only because it is available: a stable pad is a valid design.

A clear pluck

Load Pluck-Init. Use a short filter-envelope decay, low sustain and a positive filter amount. Match the amp envelope to the desired note length. Add a quiet tempo-synced delay and little reverb so the attack remains clear. Play repeated notes at the song tempo before saving.

Compare fairly

A louder setting can seem more impressive without being better. Match output levels when comparing variants. Change one section at a time, save useful variations, and use HOW IT WORKS to revisit the saved signal-path choices.

12 / Troubleshooting and support

No sound

Check host audio output, track routing and monitoring, mixer source levels and OUTPUT. Disable ARP/HOLD for a simple key test. A filter with very low cutoff and full mix can strongly attenuate the sound. Try Init or a known factory preset.

A control seems ineffective

Check the relevant enable switch and MIX. DETUNE and WIDTH have little audible effect when unison side voices are not mixed in. WIDE needs stereo content. The scope is before effects. HOW IT WORKS follows saved settings. An LFO may act through a matrix row even with its direct amount at zero.

A sound continues

Check amp release, sustain pedal, ARP HOLD, delay feedback and reverb decay. If sound continues unexpectedly after these are cleared, stop playback and reload the instrument in a test project. Report the exact preset, MIDI sequence, host and build information rather than overwriting an important project.

Preset or display issue

Clear search/category filters if a preset seems missing. Reload a user preset before Save if its association was lost after project recall. Try another GUI scale if the window does not fit. Reopen the editor after an update, and verify the About build. Imported malformed or conflicting files can be skipped; read the import result.

Project and credits

SAWSTAR: Simple Synth - Big Sound. The Sounds of Trance. Created by Gyuricza Róbert / RobCZart82. Project, source and issue reporting: https://github.com/RobCZart82/SAWSTAR . The project license and third-party notices are supplied in the repository. See the release notes for download packages and supported configurations.